Backbone-preserving peptide substitution and potential-guided refinement of TCR–peptide–MHC structures

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tcren technical appendix

Abstract

`tcren` ranks candidate epitopes for a fixed TCR–peptide–MHC structure by summing a residue-level pairwise statistical potential over the observed contact map [2]; because the potential depends only on residue identity and the closest-atom contact geometry, a candidate is scored *virtually*—its amino acids are threaded onto the native contacts with no atom moved. This appendix documents two coordinate-level extensions that the virtual score does not need but downstream re-docking does: (i) a backbone-preserving *peptide substitution* that produces the coordinates of a new equal-length epitope on the native scaffold, and (ii) a knowledge-based *potential-guided refinement* of the peptide pose—a statistical contact potential plus a soft heavy-atom clash term and a harmonic restraint to the input pose, sampled by rigid-body Metropolis Monte Carlo [5, 3] and implemented as a `pybind11`/C++ kernel. We give the energy function, prove that the restraint is necessary (without it the rigid-body minimiser is unbounded and ejects the peptide), and state the sampler and its cost. The method is knowledge-based, in the spirit of contact potentials of mean force [9, 6]; it is *not* a physics force field. Physics-grade relaxation is delegated to Rosetta FlexPepDock [7, 4].

1. Scope

For a chain-typed complex `tcren` builds the inter-chain contact map—closest heavy-atom pairs at a 5 Å cutoff, reduced to one pair per residue pair—and scores a candidate peptide s on the TCR↔peptide interface as $\sum_{(i,j)} \phi(a_i, s_{\pi(j)})$, where ϕ is the TCRen statistical potential, a_i the fixed partner residue and π the contact's peptide position [2]. This is a fixed-backbone, fixed-contact-set approximation: the native pose stands in for every candidate. It is fast (microseconds per peptide) and is the basis of all ranking in `tcren`. The two operations below relax that approximation when a candidate's *geometry*—not just its score on the native contacts—is required, e.g. to seed re-docking, to relieve a clash introduced by substitution, or to compare poses across candidates.

2. Backbone-preserving peptide substitution

Definition 1 (Substitution). Let the native peptide chain have residues $r_1, \dots, r_L$ with heavy-atom coordinate sets $X_1, \dots, X_L$, and let $s \in \mathcal{A}^L$ be a new sequence over the twenty amino acids. The

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substitution sub(· , s) returns a chain whose i-th residue has identity s_i and retains exactly the backbone atoms $\{\mathrm{N}, \mathrm{C}_\alpha, \mathrm{C}, \mathrm{O}\}$ together with C_β when $s_i \neq$ Gly, with all coordinates inherited from X_i; side-chain atoms beyond C_β are discarded.

Rationale. A residue-level potential reads the contact map through residue *identity* and the closest-atom (or C_β) distance [6]; the backbone $+\,\mathrm{C}_\beta$ frame therefore carries everything the score and the refinement energy of § 3 consume, while the native backbone—rigidly constrained between the MHC groove walls and the TCR loops—is the appropriate scaffold for an equal-length epitope. Rebuilding the discarded side chains (a backbone-dependent rotamer search [8]) and full physics relaxation are deliberately *not* performed here; substitution only fixes identity and C_β geometry, leaving heavier modelling to the refiner or to an external tool. The map is purely combinatorial—no alignment, no optimisation—so it is exact and order-preserving on the peptide's sequential indexing.

3. Potential-guided refinement

Write $x \in \mathbb{R}^{3M}$ for the M peptide heavy-atom coordinates (the only degrees of freedom) and $y \in \mathbb{R}^{3M'}$ for the fixed partner (TCR + MHC) heavy atoms; a indexes peptide residues by amino acid, b the partner residues. Let $\phi : \mathcal{A} \times \mathcal{A} \to \mathbb{R}$ be a residue-level contact potential, signed so that favourable contacts are negative (TCRen [2], or the Miyazawa–Jernigan general contact energy [6]). With a contact cutoff r_c define the residue contact set

$$\mathcal{C}(x) \;=\; \Big\{(i,j) : \min_{p \in i,\, q \in j} \|x_p - y_q\| \le r_c\Big\}.$$

Definition 2 (Refinement energy).

$$E(x) \;=\; \underbrace{\sum_{(i,j)\in\mathcal{C}(x)} \phi(a_i, b_j)}_{\text{statistical potential}} \;+ w_{\mathrm{c}} \sum_{p,q\,:\,\|x_p - y_q\| < d_0} (d_0 - \|x_p - y_q\|)^2 \;+\; w_{\mathrm{r}} \sum_{p=1}^{M} \|x_p - x_p^0\|^2,$$

where the second term is a soft quadratic penalty on heavy-atom pairs closer than d_0 (a "soft-core" clash, default $d_0 = 3$ Å) and the third is a harmonic restraint to the input pose x^0. Defaults: $r_c = 5$ Å (the TCRen contact definition), $w_{\mathrm{c}} = w_{\mathrm{r}} = 1$.

Why the restraint is necessary. The statistical-potential term alone cannot be minimised by moving a *rigid* peptide.

Proposition 1. *Restrict x to the rigid-body family $x = Rx^0 + t$, $R \in SO(3)$, $t \in \mathbb{R}^3$. The map $(R, t) \mapsto \sum_{(i,j)\in\mathcal{C}} \phi(a_i, b_j)$ takes finitely many values and is constant on the open set of t with $\|t\|$ large enough that $\mathcal{C} = \varnothing$, where it equals 0. Hence if the native contact energy $\sum_{(i,j)\in\mathcal{C}(x^0)} \phi > 0$, every minimiser of the potential term alone has $\mathcal{C} = \varnothing$: the peptide is ejected from the pocket. The clash term bounds penetration but not ejection. Adding $w_{\mathrm{r}} \sum_p \|x_p - x_p^0\|^2$ makes E coercive ($E \to \infty$ as $\|x\| \to \infty$), so a finite minimiser near x^0 exists, and w_{r} controls its distance from x^0.*

Statistical potentials are not physical energies—a contact may be net unfavourable—so "more burial" need not lower E; the restraint converts a global, unbounded search into the intended *local* refinement.

Sampling. E is minimised by simulated-annealing Metropolis Monte Carlo [5, 3]. At step t a trial pose is proposed by a rigid move of the peptide: a Gaussian translation $\delta \sim \mathcal{N}(0, \sigma_t^2 I_3)$ and a rotation about the peptide centroid by a random axis and angle $\theta \sim \mathcal{N}(0, \sigma_r^2)$. The trial is accepted with probability $\min(1, \exp(-\Delta E/T_t))$, with the temperature cooled linearly $T_t : T_0 \downarrow T_1$; the lowest-energy pose visited is returned. The chain is deterministic given its pseudo-random seed.

Implementation and cost. Each energy evaluation is $O(MM')$; partner atoms are pre-filtered to those within a 12 Å shell of the peptide, so M' is the interface, not the whole complex, and several thousand steps complete in well under a second. The kernel is C++ (`pybind11`); the dense potential matrix is passed by buffer and indexed in $O(1)$. A native pose is a fixed point of the move set up to thermal noise (it barely moves); a clashed or perturbed input relaxes locally, trading clash for restraint.

4. Assumptions and limitations

The peptide is treated as a *rigid* body: there is no backbone flexibility and no side-chain repacking, so substitution leaves heavy atoms beyond C_β unmodelled and the refiner cannot resolve rotameric clashes (a backbone-dependent rotamer pass [8] is the natural next step). A single potential ϕ is used across all peptide–partner contacts, whereas the TCR-facing and groove-facing faces have different statistics; the clash term is a heuristic soft-core, not a Lennard–Jones repulsion. The restraint weight w_{r} is a tunable that sets the locality radius. For physically realistic, full-atom relaxation—side-chain optimisation and backbone minimisation under an all-atom force field—`tcren` defers to Rosetta FlexPepDock [7, 4] with the ref2015 energy function [1], invoked as an external process; the C++ kernel here is a fast knowledge-based pre-refinement, not a substitute for it.

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